SULIT

CP115
Computer Programming I
Semester I
Session 2025/2026
2 Hours 30 Minutes

CP115
Pengaturcaraan Komputer I
Semester I
Sesi 2025/2026
2 Jam 30 Minit

Nama	No Matrik	Kelas

KOLEJ MATRIKULASI LABUAN
LABUAN MATRICULATION LABUAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME PRE-EXAMINATION

PENGATURCARAAN KOMPUTER I
2 JAM 30 MINIT

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No Soalan	Markah	
	Pemeriksa	KP/KKP
1 (17)		
2 (15)		
3 (17)		
4 (19)		
5 (7)		
6 (10)		
Jumlah (85)		

Kertas soalan ini mengandungi 10 halaman bercetak
This question paper consists of 10 printed pages

1.

a. State and define the first and last steps in problem solving.

[4 marks]

_____

_____

_____

_____

b. For each situation below, identify whether each is Input, Process or Output.

[5 marks]

Situations	Answer
A motion sensor detects movement in a hallway.	
The irrigation system calculates fertilizer ratios based on soil data.	
A seismograph records ground vibrations.	
A student clicks the "Submit" button after completing an online quiz.	
The cash register looks up the product price and adds it to the total.	

c. For each situation below, identify the Input, Process and Output.

i. The student grade system determines if a student passes or fails a subject. The system receives the student's test score as input. It checks if the score is 50 or above. If the score is 50 or above, it outputs "PASS". If the score is below 50, it outputs "FAIL".

[4 marks]

Input	
Process	
Output	

ii. Every day, Aminah needs to calculate the daily profit for each of her 5 breakfast stalls scattered around Labuan. For each stall, she has to enter the daily sales amount and the daily stall's expenses.

[4 marks]

Input	
Process	
Output	

2.

a. Algorithm can be represented in many ways including pseudocode and flowchart. Define the meaning of flowchart.

[1 marks]

_____

_____

b. Draw the shape of the flowchart components based on the description.

[3 marks]

Description	Answer
Used for data that is entered by the user or information that is displayed to the user	
Used for calculations or assignments	
Used for making choices	

c. Convert the pseudocode below into a flowchart.

[5 marks]

```
START
        Enter age
        if age >= 18
                Print "You can vote"
        else
                Print "You cannot vote"
        end if
END
```

d. Aminah owns a small grocery store and needs to calculate her total daily sales. Write a pseudocode that processes sales for 10 customers throughout the day. For each customer, the system gets the purchase amount and adds it to the total sales. After serving all 10 customers, display the total daily sales amount.

[6 marks]

3.

a. Programming languages are used to build software, games, websites, and more. Define what a programming language is.

[1 marks]

_____

_____

b. Give one (1) advantage and one (1) disadvantage of using machine language.

[2 marks]

Advantage	Disadvantage

c. Determine whether each of the following identifiers is valid or invalid.

[4 marks]

Identifier	Answer
Is_valid:	
_____value	
1_variable	
number-1	

d. For the Python code below, a user executes the program and enters 5 as the input. Identify which line will throw an error and what type of error it is.

[2 marks]

```
1.  value = input("Enter a number")
2.  if value > 0:
3.      value = value + 5
4.      print(value)
```

Line of code	Type of error

e. State the output of the Python program below.

[4 marks]

```
1.  value = "Hello World"
2.  print(len(value))
3.  print(len(value) // 5)
4   print(value.upper())
5.  print(f"Length is {len(value)}")
```


f. State two (2) purposes of using comments in Python.

[2 marks]

_____

_____

g. Convert the following arithmetic expressions into Python statements.

[2 marks]

Arithmetic expression	Python statements
$y = 2^{3+r} + 5y^{2}$	
$x = \frac{k+2+r}{gk}$	

4.

a. Compare the usage of the below Python keywords for control flow statements.

[2 marks]

break	*continue*

b. Ali has written the Python code below that allows him to enter the temperature of his garden and the program will print the temperature. The program will

continually run until he enters any value below 0. Identify one (1) component of the sentinel loop that he did not add to his code. [1 mark]

```
1.  temperature = float(input("Enter the temperature")
2.  while temperature > 0 :
3.      print("Your garden temperature is ", temperature)
```

_____

_____

c. Convert the below Python statements using *for* statements.

[4 marks]

```
1.  i = 101
2.  while i > 0 :
3.      print(i)
4.      i = i - 5
5.  print(i)
```

d. Give examples of code segments for each of the following selection control structures.

[3 marks]

Type of Selection Control Structures	Code Segment
Single selection	
Dual Selection	
Multiple Selection	

e. Evaluate the Python statements below and find the value of each variable.

[4 marks]

Python statements	Answer
y = 8 % 3 = = 2 and 10 > 7	y =
x = not (5 < 3) or 4 = = 4	x =
z = (15 - 3) / 4 = = 3.0	z =
a = 6 + 4 = = 10 and 5 > 3	a =

f. Convert the below pseudocode into a Python program.

[5 marks]

```
1.  START
2.      a = 5
3.      While a < 15
4.          If a = = 10
5.              Print a
6.          a = a + 1
7.          end if
8.      end while
9.  END
```

5. Write a Python program that will continually accept students' marks ranging from 0 - 100. The program will terminate if the user enters any value outside of that range. Assume all the input user will enter is only numbers. The program will display the highest marks and the average marks. *(Use test pad or back page)

[7 marks]

6. Write a Python program that processes exam scores for 37 students in a class. For each student, the program should ask for their name and exam score. The program should count how many students passed (score 50 or above) and how many students failed (score below 50). At the end, display the total number of students who passed, the total number who failed, and the class average score. Assume for exam score, the user will only enter value ranging from 0 - 100. *(Use test pad or back page3)

[10 marks]

END OF QUESTION PAPER